Nila Mönig, PhD

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PROFILE

PhD neuroscientist with 10+ years connecting biology, computation and engineering. Develops deep-learning pipelines for large, complex image data, plus the tools, quality control, documentation and partnerships that make them usable. Hands-on from hardware to code: builds solutions where no standard tool exists.

CORE STRENGTHS

- **Computational imaging and machine learning:** deep-learning segmentation of TB-scale EM data on HPC.
- **Bridging disciplines and partners:** 10+ years in interdisciplinary teams and external partnerships.
- **Building tools and hardware:** napari and Neuroglancer tools, self-built behavioural apparatus.
- **Communication and knowledge transfer:** advises colleagues; thesis, publications, preprints, posters, talks.

CORE COMPETENCES

Computational microscopy and image analysis: Electron microscopy | Fluorescence microscopy | Image analysis | Computer vision | Image registration | Segmentation | Object classification | Feature extraction
Machine learning & AI: PyTorch | PyTorch Lightning | MONAI | Gunpowder | scikit-learn | Weights & Biases | U-Net / ResNet | Ground-truth generation | Model evaluation
Programming & tools: Python | MATLAB | Git | napari | Neuroglancer | BigWarp | FIJI/ImageJ | ilastik | scikit-image | NetworkX | BigQuery
Scalable data processing: SLURM/HPC | Array-based parallel deployment and inference | Process automation | Data integration
Workflow & reproducibility: Automation | Code versioning | Documentation | Quality control | Annotation quality assessment | Benchmarking
Data analytics: Statistical analysis | Multi-dimensional data analysis | Clustering | Dimensionality reduction
Collaboration & communication: Cross-disciplinary collaboration | Annotation-partner coordination | Mentoring and knowledge transfer | Scientific writing | Presentation | Figure and poster design (Illustrator)
Hardware and prototyping: Prototyping | Fabrication | Fluidics | Control electronics | Soldering
Experimental methods: Zebrafish behaviour | Electrophysiology | Pharmacology | Immunohistochemistry | Neuron tracing | Cell culture

PROFESSIONAL EXPERIENCE

Independent Work
Jul 2026 – present

- Finalising the synapse-detection pipeline (code cleanup, documentation, validation) for release
- Revising a manuscript for journal publication

POSTDOCTORAL RESEARCH SCIENTIST
Friedrich Miescher Institute for Biomedical Research, Basel (lab of Rainer W. Friedrich) – Oct 2022 – Jun 2026

Led development of scalable data pipelines for connectomics analysis in zebrafish neuroscience.

- **Pipeline development:** designed and implemented, as sole developer, a high-throughput synapse-detection pipeline for tissue with non-standard properties.
- **Deep learning on HPC:** trained, evaluated and deployed segmentation and classification models (U-Net, ResNet variants) for TB-scale EM data with array-based parallel inference.
- **Tool development:** created and maintained proofreading and training-data tools in napari and Neuroglancer, with documentation and user feedback, in collaboration with imaging-facility specialists.
- **Partner coordination:** set up and coordinated a high-throughput proofreading partnership with an external company; defined annotation requirements and best practices, ran quality assessment with structured error feedback, and wrote interface scripts to reconcile platforms and data formats.
- **Segmentation and ground truth:** took over responsibility, on our side of a Google collaboration, for neuron segmentation of a large multimodal dataset of highly variable quality: inspected model output, judged adequacy, masked data, generated and corrected ground truth; contributed to cell mapping (BigWarp).

- **Strategy and data quality:** designed the experimental workflow in own projects and the computational strategy of the synapse pipeline; gave acquisition teams image-quality feedback for segmentation.
- **Knowledge transfer:** advised colleagues on segmenting their own datasets (timelines, steps, ground truth) and on connectomics methods; accompanied their contact with the proofreading company; taught annotation and tool use; supervised an intern on image registration.
- **Documentation and writing:** wrote tool documentation and a detailed guideline for synapse annotation.

PHD CANDIDATE

Friedrich Miescher Institute for Biomedical Research, Basel (lab of Rainer W. Friedrich) – May 2012 – Sep 2022

Thesis: Neuronal Organisation of the Olfactory Bulb in Adult Zebrafish | magna cum laude (5.5/6.0)

Developed integrated connectomics and behavioural-analysis workflows to study the zebrafish olfactory system.

- **Collaboration with Google Research:** partnered on neuron segmentation for TB-scale EM data; assessed model output by visual inspection, reported errors and generated targeted ground-truth data to close gaps.
- **Ground truth and data quality:** generated ground-truth training datasets, gaining a deep understanding of model requirements and data quality.
- **Proofreading tools:** developed a Neuroglancer-based interface for inspecting and curating automated segmentations (15–50× faster reconstruction) and a tool for manual neuron classification.
- **Analysis:** proofread >450 neurons; established a morphological classification scheme; classified cells automatically by clustering skeleton-based morphological and volumetric features.
- **Process automation:** co-developed and automated odour-discrimination assays for adult zebrafish, cutting manual work by 87% while doubling throughput; basis of three peer-reviewed publications.
- **Data analysis:** built a multi-dimensional analysis pipeline (Python/MATLAB) for behavioural video features.
- **Hardware:** designed and built behavioural apparatus (prototyping, fabrication, fluidics, control electronics).
- **Cross-disciplinary projects:** contributed behavioural training and analysis to neuro-behavioural projects.
- **Mentoring:** trained a student assistant in neuron proofreading.

RESEARCH ASSISTANT, University of Freiburg – Jan 2011 – Mar 2012

- **Prepared and maintained** primary neuronal cell cultures.
- **Implemented** closed-loop feedback stimulation circuits (Tucker Davis Technologies); performed electrophysiology, pharmacology, immunohistochemistry and neuron tracing.

STUDENT ASSISTANT, University of Freiburg – Jun 2007 – Sep 2009

- **Teaching assistant** for an introductory MATLAB course.
- **Contributed** to data-management system development and neuroanatomical analyses.

EDUCATION

- PhD in Neuroscience, University of Basel (2022) – magna cum laude (5.5/6.0, Swiss scale)
- Diplom in Biology (≈ MSc), University of Freiburg (2009) – 1.0 (best possible on the German scale)

SELECTED PUBLICATIONS & PRESENTATIONS

- Mönig N.R.*, Januszewski M.* et al., 2026, **eLife** Reviewed Preprint 112715
- Montaño Crespo R.E., Mönig N.R. et al., 2026, **bioRxiv** 2026.09.18.752720 (preprint)
- Frank T., Mönig N.R., Satou C. et al., **Nat Neurosci** 22: 1844–1856 (2019)
- Namekawa I.*, Mönig N.R.*, Friedrich R.W. **Exp Brain Res** 236: 2959–2969 (2018) *(*equal contribution)*

Posters: Connectomics (2019, 2022, 2024), ECRO (2022), Zebrafish Brain Conference (2022), FENS (2018). Talks at institute and regional meetings.

SELECTED COURSES & TRAINING

- Entrepreneurship Training, Innosuisse Module 2 (Innovation Office, University of Basel, 2026)
- Advanced Python and Machine Learning (University of Basel, 2024)
- Deep Learning for Image Analysis (EMBL, Heidelberg, 2023)

LANGUAGES

German (native), English (C2), French (B1)

INTERESTS

Volunteer conservation work, running, healthy eating and cooking, furniture upcycling, knitting